

Algorithmic Pseudocode Sample 44

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

- 1: Initialize the weights of an RL agent and a Self-Explainer (SE-Net)
 - 2: Initialize buffers $\mathcal{D}_{success}$ and $\mathcal{D}_{failure}$ for training the Self-Explainer and D_{RL} for RL from Demonstrations as in the RLfD works [?, ?]
 - 3: Store expert experiences into $\mathcal{D}_{success}$ and D_{RL} . Pretrain the RL agent with experiences sampled from D_{RL}
 - 4: Sample K trajectories with a random policy and add them to D_{RL} . Also add successful and unsuccessful trajectories to $\mathcal{D}_{success}$ and $\mathcal{D}_{failure}$ respectively.
 - 5: **for** $episode = 1; episode \leq N; episode++$ **do**
 - 6: Sample experiences (including grounded predicate values) from $\mathcal{D}_{success}$ and $\mathcal{D}_{failure}$ $\triangleright$
 Train Self-Explainer
 - 7: Compute utility weights u and shaped reward prediction $\hat{r}_\theta(s_t, u_t, a_t, s_{t+1}, u_{t+1})$ by using SE-Net, Eqs. ?? and ??
 - 8: Update the SE-Net via binary cross entropy loss L_{SE} (Eq. ??) to distinguish successful experiences from unsuccessful experiences
 - 9: Sample experiences with grounded predicate values from D_{RL} $\triangleright$ **Train RL agent**
 - 10: Run SE-Net on sampled states to obtain utility weights u
 - 11: Augment input states with grounded predicate values and utility weight values
 - 12: Augment rewards with predicted shaped reward by using Eq. ??
 - 13: Update RL agent with the augmented experiences
 - 14: Use RL agent to sample a new trajectory and add it to D_{RL} . If the trajectory is successful, it would also be added to $\mathcal{D}_{success}$. Otherwise, it would also be added to $\mathcal{D}_{failure}$ $\triangleright$ **Sample a new trajectory**
 - 15: **end for**
 - 16: **return** a trained RL agent and SE-Net
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